

BHARATH NAVEEN

Data Science · Machine Learning · LLM Evaluation · Python · SQL · Docker · AWS

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SUMMARY

Data scientist and machine learning engineer who builds end-to-end systems, from an ambiguous problem to a containerized, tested application. Built a 5-layer phishing detection system with 4 ML models and explainable verdicts, backed by 385 automated tests. Currently designing LLM reasoning evaluations with Handshake AI, with three years of enterprise data and analytics experience at TCS.

EXPERIENCE

AI Training & Evaluation Contractor | Handshake AI (Contract, Remote) Sep 2026 – Present

- Design fair, expert-level evaluation challenges that test the reasoning of large language models, targeting known reasoning weaknesses and validated through blind testing so the checks never see the answer key.
- Run ablation, sensitivity, and controlled one-variable comparisons to isolate what drives an outcome, and catch and fix leakage in my own test setups.
- Write structured rationales that separate genuine model errors from ambiguous specifications, giving reviewers clear, actionable feedback.

System Engineer, Data and Analytics | Tata Consultancy Services Apr 2023 – Aug 2024

- Automated SQL querying and data-collection workflows in Python, saving about 3 days of manual effort per reporting cycle and improving data quality and consistency.
- Built 2 self-refreshing Power BI dashboards (bug tracking and feature delivery) for Microsoft stakeholders, replacing manual update cycles with live reporting.
- Analyzed 300+ test and 180+ defect records per cycle to flag high-risk areas and surface actionable insights for engineering leads.

Associate System Engineer | Tata Consultancy Services Jul 2022 – Apr 2023

- Led 2 analysts on data-collection and validation workflows, building automation that reduced manual collection effort and sped up defect-pattern analysis.

Assistant System Engineer, Trainee | Tata Consultancy Services Apr 2021 – Jun 2022

- Validated data integrity across 300+ structured checks in enterprise Microsoft-product environments, building the foundation for later analytics and reporting work.

KEY PROJECTS

Phishing Detection System | MS Capstone, University of Arizona 2026

- Built a 5-layer phishing detection system (XGBoost URL model, Playwright page capture, rule-based evidence judge) on a 795K-URL dataset; F1 0.80 and ROC-AUC 0.88 on 147.7K test URLs from websites never seen in training.
- Audited my own first version, found train/test leakage (55% of test rows shared websites with training), and rebuilt the pipeline with domain-grouped splits, leakage guards and a one-command reproducible evaluation.
- Cut false alarms on real brand sites from 40.4% (URL model alone) to 2.9% (4 of 136) by layering page evidence and adjudication rules, tuned on a validation split under a plan written before tuning.

Cohort Analytics Platform, Backend Prototype | University of Arizona, Vertically Integrated Projects Spring 2026

- Demonstrated single-plan revert as an alternative to whole-table backup restores, in a prototype backend for a university degree-progress platform: an AWS Lambda function that stores every degree plan as a version (DynamoDB for the current plan, S3 for history) and rolls back to the last school-approved one, built and tested without touching the live database.

BN Motors SQL Data System | Personal Project 2025

- Designed a normalized 25-table MySQL dealership schema from the ER model up, with 10 analytical queries (window functions, CTEs, subqueries) and 3 reporting views for inventory aging, gross profit, and customer lifetime value.

Loan Default Prediction | Academic Project 2025

- Benchmarked 5 models on German Credit data (1,000 borrowers, 26% default rate), tuned the top 2 for F1, and reached 0.77 ROC-AUC with Gradient Boosting; top drivers were employment duration and checking-account status.

SKILLS

Languages: Python, SQL

ML / AI: scikit-learn, XGBoost, LightGBM, Random Forest, Gradient Boosting, SVM, Logistic Regression

Evaluation: LLM and model evaluation, ablation analysis, A/B testing, data leakage prevention, adversarial evaluation (red-teaming)

Data and Viz: pandas, NumPy, Playwright, BeautifulSoup, Streamlit, Power BI

Cloud and Infra: Docker, AWS (Lambda, DynamoDB, S3), Git, CI/CD, pytest, MySQL

EDUCATION

MS, Information Science (Machine Learning) | University of Arizona | GPA 3.89 Jul 2024 – May 2026

BE, Mechanical Engineering | SEA College of Engineering and Technology 2016 – 2020

Certifications | Microsoft Certified: Azure Fundamentals; Microsoft Office Specialist: Excel Associate, Word Expert, PowerPoint Associate